

COMPUTER NETWORKS

Part 2

Complete Study Notes

Unit 3 MAC Sublayer and IP Addressing

Unit 4 Routing and IP Header

UNIT 3 — MAC Sublayer and IP Addressing

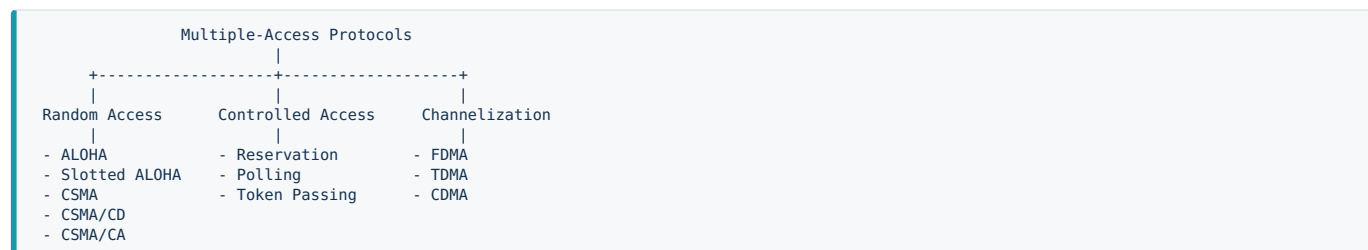
PART A — The MAC Sublayer (Medium Access Control)

3.1 The Need for Media Access Control

When multiple devices share a single transmission medium (a **broadcast link** or **multipoint link**), we need a protocol to coordinate access — otherwise frames from two stations will overlap and both will be destroyed. This overlap is called a **collision**.

The MAC sublayer's job is to decide **which station may transmit and when**.

Classification of Multiple-Access Protocols



3.2 Random Access (Contention) Protocols

No station is superior; no station controls another. Each station transmits when it has data, and there is contention for the channel.

a) Pure ALOHA

Developed at the University of Hawaii (1970s) for a wireless packet network.

- A station transmits **whenever it has data**.
- If frames collide, both are destroyed.
- The sender waits for an ACK; if no ACK arrives within a timeout, it assumes a collision, waits a random **back-off time** (**T_B**) and resends.
- After **K_{max}** attempts, the station gives up.

Vulnerable time = $2 \times T_{fr}$ (twice the frame transmission time).

Throughput: $S = G \times e^{(-2G)}$, maximum **S = 0.184 (18.4%)** at **G = 0.5**.

b) Slotted ALOHA

Improves Pure ALOHA by dividing time into **slots** of T_{fr} seconds. A station may transmit **only at the beginning of a slot**.

Vulnerable time = T_{fr} .

Throughput: $S = G \times e^{(-G)}$, maximum **S = 0.368 (36.8%)** at **G = 1**.

c) CSMA — Carrier Sense Multiple Access

Principle: "**Sense before transmit**" / "listen before talk". A station first senses whether the medium is busy.

Collisions can still occur because of **propagation delay** — a station may start sending before another station's signal reaches it.

Persistence Methods:

Method	Behavior when channel is busy	Behavior when idle	Characteristic
1-persistent	Keep sensing continuously	Transmit immediately (probability 1)	Highest chance of collision; used in Ethernet
Non-persistent	Wait a random time, then sense again	Transmit immediately	Reduces collisions but reduces efficiency (channel may sit idle)
p-persistent	Continue sensing (used with slotted channels)	Transmit with probability p ; with probability $1-p$, wait for the next slot	Best of both worlds; reduces collisions and improves efficiency

d) CSMA/CD — Carrier Sense Multiple Access with Collision Detection

Used in **wired Ethernet (IEEE 802.3)**.

Procedure: 1. Sense the channel; if busy, wait (per persistence method). 2. Transmit the frame while **simultaneously monitoring** the channel. 3. If a collision is detected: - Immediately **abort** the transmission. - Send a short **jam signal** (48 bits) so all stations know a collision occurred. - Increment the attempt counter K . - Wait a random back-off time using **binary exponential back-off**: choose a random number R in the range 0 to $2^K - 1$, and wait $R \times \text{slot time}$. - Retry (up to 15 attempts, then abort).

Minimum Frame Size requirement: A frame's transmission time must be at least $2 \times T_p$ (twice the maximum propagation time) so that a sender is still transmitting when the collision signal reaches it. This is why standard Ethernet has a **minimum frame size of 64 bytes (512 bits)** and a maximum segment length of 2500 m.

e) CSMA/CA — CSMA with Collision Avoidance

Used in **wireless networks (IEEE 802.11 Wi-Fi)**, because in wireless: - A station cannot transmit and receive at the same time (cannot detect its own collision). - **Hidden terminal problem** and **exposed terminal problem** exist.

So collisions are **avoided** rather than detected:

1. **Interframe Space (IFS)** — when the channel is found idle, the station waits a period of time (IFS) before sending. Different priorities get different IFS durations ($\text{SIFS} < \text{PIFS} < \text{DIFS}$).
2. **Contention Window** — an amount of time divided into slots. The station picks a random number of slots to wait; the window size doubles on each failure (binary exponential back-off).
3. **Acknowledgement** — positive ACK plus timeout guarantees delivery.
4. **RTS/CTS Handshake (optional)** — the sender transmits a **Request To Send**; the receiver replies with **Clear To Send**. All other stations hearing either frame set their **NAV (Network Allocation Vector)** and stay silent. This solves the hidden terminal problem.

3.3 Controlled Access Protocols

Stations consult one another to decide who may send. No collisions occur.

a) Reservation

Time is divided into intervals. Each interval begins with a **reservation frame** containing one minislot per station. A station that wants to send sets its bit in its minislot. Then stations transmit in the order of their reservations.

b) Polling

Works with a **primary** station controlling several **secondary** stations. All exchanges go through the primary. - **Poll** — primary asks a secondary if it has data to send. - **Select** — primary tells a secondary it is about to send data to it. - **Drawback:** Polling overhead; single point of failure at the primary.

c) Token Passing

Stations are organized in a logical **ring**. A special frame called a **token** circulates. Only the station holding the token may transmit. After sending (or if it has nothing to send), it passes the token to its logical successor. - Needs **token management**: token loss detection, duplicate token handling, and a **token holding time** limit. - Used in **Token Ring (IEEE 802.5)** and **FDDI**.

3.4 Channelization

The available bandwidth is shared in time, frequency, or through code.

Method	How the channel is shared	Used in
FDMA (Frequency Division Multiple Access)	Bandwidth divided into frequency bands; each station gets a band for the whole time	1G analog cellular (AMPS)
TDMA (Time Division Multiple Access)	Each station gets a time slot; all use the whole bandwidth during their slot	2G GSM
CDMA (Code Division Multiple Access)	All stations transmit simultaneously on the whole bandwidth, separated by unique orthogonal chip codes (Walsh codes)	3G, IS-95

Note: FDMA/TDMA/CDMA are *access* methods (multiple stations sharing a link); FDM/TDM/CDM are *multiplexing* techniques (one device combining multiple streams).

3.5 Wired LANs — Ethernet

IEEE 802 Standards

The IEEE Project 802 divides the data link layer into **LLC (802.2)** and **MAC** sublayers.

Standard	Technology
IEEE 802.3	Ethernet (CSMA/CD)
IEEE 802.4	Token Bus
IEEE 802.5	Token Ring
IEEE 802.11	Wireless LAN (Wi-Fi)
IEEE 802.15	Wireless PAN (Bluetooth)
IEEE 802.16	Broadband Wireless (WiMAX)

Ethernet Frame Format (IEEE 802.3)

Field	Size	Purpose
Preamble	7 bytes	Alternating 1s and 0s for receiver synchronization (added by the physical layer)
SFD (Start Frame Delimiter)	1 byte	10101011 — signals the start of the frame
Destination Address (DA)	6 bytes	MAC address of the destination
Source Address (SA)	6 bytes	MAC address of the sender
Length / Type	2 bytes	Number of bytes in data field, or the upper-layer protocol type
Data / Payload	46-1500 bytes	The encapsulated packet (padded if less than 46)
CRC / FCS	4 bytes	CRC-32 error detection

- **Minimum frame length:** 64 bytes (512 bits) — required by CSMA/CD.
- **Maximum frame length:** 1518 bytes — limits memory needed and prevents one station monopolizing the medium.

MAC Addresses

- **48-bit (6-byte)** physical address, usually written as 12 hexadecimal digits: `06:01:02:01:2C:4B`
- Burned into the NIC (also called a "burned-in address" / hardware address).
- First 24 bits = **OUI (Organizationally Unique Identifier)**, assigned to the manufacturer.
- **Unicast** — the least significant bit of the first byte is 0.
- **Multicast** — the least significant bit of the first byte is 1.
- **Broadcast** — all 48 bits are 1: `FF:FF:FF:FF:FF:FF`

Ethernet Evolution

Generation	Speed	Standards / Media
Standard Ethernet	10 Mbps	10Base5 (Thick coax), 10Base2 (Thin coax), 10Base-T (UTP), 10Base-F (Fiber)
Fast Ethernet	100 Mbps	100Base-TX, 100Base-FX, 100Base-T4
Gigabit Ethernet	1 Gbps	1000Base-T, 1000Base-SX, 1000Base-LX, 1000Base-CX
10 Gigabit Ethernet	10 Gbps	10GBase-S, 10GBase-L, 10GBase-E — full duplex only, no CSMA/CD needed

Naming convention: **10Base-T** = 10 Mbps, Baseband signaling, Twisted-pair.

Bridged, Switched, and Full-Duplex Ethernet

- **Bridged Ethernet** — bridges divide a LAN into separate **collision domains**, raising bandwidth per domain and isolating traffic.
- **Switched Ethernet** — a switch is essentially a multiport bridge; each port is its own collision domain.
- **Full-Duplex Ethernet** — with a dedicated switch port and separate transmit/receive paths, there are **no collisions**, so CSMA/CD is turned off and capacity doubles.

Connecting Devices Summary

Device	Layer	Function
Repeater	Physical	Regenerates the signal to extend distance
Hub	Physical	Multiport repeater; one collision domain, one broadcast domain
Bridge	Data Link	Filters traffic using MAC addresses; separates collision domains
Switch	Data Link (L2)	Multiport bridge; each port = a collision domain; builds a MAC address table
Router	Network	Connects different networks; routes by IP address; separates broadcast domains
Gateway	All layers	Connects networks with different architectures/protocols

PART B — IP Addressing

3.6 What is an IP Address?

An **IP address** is a **logical address** used at the network layer to uniquely identify the connection of a device (host or router) to the Internet.

- **IPv4** — 32 bits, giving $2^{32} \approx$ **4.29 billion** addresses.
- Addresses are **unique** and **universal**.
- A device with multiple network connections (a router) has **multiple IP addresses**, one per interface.

Address Notations

1. **Binary notation:** 01110101 10010101 00011101 00000010
2. **Dotted-decimal notation:** 117.149.29.2 — four decimal numbers (0–255) separated by dots.
3. **Hexadecimal notation:** 0x75951D02

3.7 Classful Addressing (Legacy)

The IPv4 address space was originally divided into five classes.

Class	First bits	First byte range	Default Mask	Networks	Hosts per network
A	0	0 – 127	255.0.0.0 (/8)	128 (2 ⁷)	16,777,214 (2 ²⁴ – 2)
B	10	128 – 191	255.255.0.0 (/16)	16,384 (2 ¹⁴)	65,534 (2 ¹⁶ – 2)
C	110	192 – 223	255.255.255.0 (/24)	2,097,152 (2 ²¹)	254 (2 ⁸ – 2)
D	1110	224 – 239	—	Multicast	—
E	1111	240 – 255	—	Reserved / experimental	—

Netid and Hostid

- Class A: 1 byte netid, 3 bytes hostid
- Class B: 2 bytes netid, 2 bytes hostid
- Class C: 3 bytes netid, 1 byte hostid

Problems with Classful Addressing

- **Address depletion** — classes were too coarse. A company needing 300 hosts had to take a Class B (65,534 addresses), wasting most of them.
- Huge **internal fragmentation** of the address space.
- This led to **subnetting**, **supernetting**, and eventually **CIDR**.

3.8 Special IP Addresses

Address	Meaning
Network address (hostid all 0s)	Identifies the network itself, e.g. 192.168.1.0
Directed broadcast (hostid all 1s)	Sent to all hosts on that network, e.g. 192.168.1.255
Limited broadcast 255.255.255.255	Broadcast within the local network only
This host 0.0.0.0	Used as a source when a host doesn't yet know its address (DHCP)
Loopback 127.0.0.0/8 (e.g. 127.0.0.1)	Testing software on the local machine; packet never leaves the host
Link-local (APIPA) 169.254.0.0/16	Self-assigned when DHCP fails

Private IP Address Ranges (RFC 1918)

Not routable on the public Internet; used with NAT.

Class	Range	CIDR
A	10.0.0.0 – 10.255.255.255	10.0.0.0/8
B	172.16.0.0 – 172.31.255.255	172.16.0.0/12
C	192.168.0.0 – 192.168.255.255	192.168.0.0/16

3.9 Classless Addressing and CIDR

CIDR (Classless Inter-Domain Routing) removes the rigid class boundaries. Addresses are assigned in variable-length

blocks.

Slash Notation (CIDR Notation)

x.y.z.w / n — where n is the **prefix length**, the number of bits in the network portion.

Rules for a valid block: 1. Addresses must be **contiguous**. 2. Number of addresses must be a **power of 2**. 3. The **first address must be divisible** by the number of addresses in the block.

Key Formulas

Number of addresses in a block = $2^{(32 - n)}$ **Number of usable hosts = $2^{(32 - n)} - 2$** (excluding network and broadcast addresses) **First address (Network address) = Any address AND Mask** **Last address (Broadcast address) = First address OR (complement of Mask)**

Subnet Mask Reference Table

Prefix	Subnet Mask	Total Addresses	Usable Hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2
/16	255.255.0.0	65,536	65,534
/8	255.0.0.0	16,777,216	16,777,214

3.10 Subnetting — Worked Examples

Subnetting divides one large network into smaller logical sub-networks. It improves performance (smaller broadcast domains), security, and address efficiency.

Example 1

Given: An address 192.168.10.75 / 26 . Find the network address, broadcast address, and host range.

- Mask /26 → 255.255.255.192
- Block size = $2^{(32-26)} = 64$ addresses
- Subnets in the last octet: 0, 64, 128, 192
- 75 falls in the block starting at 64

	Value
Network address	192.168.10.64
First usable host	192.168.10.65
Last usable host	192.168.10.126
Broadcast address	192.168.10.127
Usable hosts	62

Example 2

Given: An organization is granted 190.100.0.0/16 and needs 3 subnets: one with 32,768 addresses, one with 16,384, and one with 8,192.

Solution (VLSM — Variable Length Subnet Masking):

Subnet	Addresses needed	Prefix	Range
Subnet 1	$32,768 = 2^{15}$	/17	190.100.0.0 - 190.100.127.255
Subnet 2	$16,384 = 2^{14}$	/18	190.100.128.0 - 190.100.191.255
Subnet 3	$8,192 = 2^{13}$	/19	190.100.192.0 - 190.100.223.255
Remaining	8,192	/19	190.100.224.0 - 190.100.255.255

Example 3 — Finding the block from an address

Given: 167.199.170.82/27

- Block size = $2^{(32-27)} = 32$
- $82 \div 32 = 2$ remainder 18 → block starts at $2 \times 32 = 64$
- Network address = **167.199.170.64**
- Broadcast = $64 + 32 - 1 = 167.199.170.95$
- Host range = .65 to .94 (30 usable hosts)

3.11 Supernetting (Route Aggregation)

The opposite of subnetting: combining several contiguous smaller blocks into one larger block, advertised with a **shorter prefix**. This reduces the size of routing tables.

Example: Four Class C networks `200.1.0.0/24`, `200.1.1.0/24`, `200.1.2.0/24`, `200.1.3.0/24` can be aggregated into a single `200.1.0.0/22`.

3.12 NAT — Network Address Translation

NAT allows a private network to use private IP addresses internally while sharing one (or a few) public IP addresses for Internet access.

How it works: A NAT-enabled router replaces the private source IP in outgoing packets with its public IP, records the mapping in a **translation table**, and reverses the translation on incoming replies.

Types: - **Static NAT** — one-to-one fixed mapping. - **Dynamic NAT** — maps to an address from a pool. - **PAT (Port Address Translation / NAT Overload)** — many private addresses map to one public address, distinguished by **port numbers**. This is what home routers do.

Advantages	Disadvantages
Conserves public IPv4 addresses	Breaks true end-to-end connectivity
Adds a layer of security (hides internal hosts)	Complicates protocols that embed IPs (FTP, SIP)
Easy to change ISPs	Adds processing overhead and latency

3.13 IPv6 Addressing

Created to solve IPv4 address exhaustion and add improvements.

Features

- **128-bit** address → $2^{128} \approx 3.4 \times 10^{38}$ addresses.
- Written in **hexadecimal colon notation**: 8 groups of 4 hex digits. `FDEC:0074:0000:0000:0000:B0FF:0000:FFFF`
- **Abbreviation**: leading zeros in a group may be dropped; one run of consecutive all-zero groups may be replaced by `::` (only once). → `FDEC:74::B0FF:0:FFFF`

Address Types

- **Unicast** — a single interface.
- **Multicast** — a group of interfaces; delivered to all.
- **Anycast** — a group of interfaces; delivered to the **nearest** one.
- (*IPv6 has no broadcast; multicast replaces it.*)

Advantages of IPv6 over IPv4

Feature	IPv4	IPv6
Address size	32 bits	128 bits
Header	Variable (20–60 bytes), complex	Fixed 40 bytes, simplified
Checksum	Present in header	Removed (handled by other layers)
Fragmentation	By sender and routers	Only by the sender
Configuration	Manual or DHCP	Stateless auto-configuration (SLAAC)
Security	Optional (IPSec add-on)	IPSec built in
QoS	Type of Service field	Flow Label + Traffic Class
Broadcast	Yes	No — replaced by multicast

Transition Strategies from IPv4 to IPv6

1. **Dual Stack** — a host runs both IPv4 and IPv6 simultaneously.
2. **Tunneling** — IPv6 packets are encapsulated inside IPv4 packets to cross an IPv4-only region.
3. **Header Translation** — headers are converted when one side is IPv6-only and the other IPv4-only.

UNIT 4 — Routing and IP Header

4.1 The Network Layer — Responsibilities

- **Logical addressing** (IP addresses)
- **Routing** — determining the best path
- **Packetizing** — encapsulating payload into datagrams
- **Fragmentation and reassembly**
- **Error and congestion reporting** (via ICMP)

Services: The Internet's network layer is **connectionless** and offers **best-effort delivery** — no guarantee of delivery, order, or error-free transfer. Reliability, if needed, is provided by the transport layer.

4.2 The IPv4 Header — Field by Field

The IPv4 header is **20 to 60 bytes** long (20 bytes fixed + up to 40 bytes of options).

```
 0           1           2           3
0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|Version|  IHL  |Type of Service|          Total Length         |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|          Identification        |Flags|      Fragment Offset  |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|  Time to Live |    Protocol   |     Header Checksum          |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                       Source IP Address                      |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                    Destination IP Address                     |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                    Options (if any)                          |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

Field Descriptions

Field	Size	Description
Version (VER)	4 bits	IP version — value 4 for IPv4
HLEN / IHL	4 bits	Header length in 4-byte words . Min value 5 (20 bytes), max 15 (60 bytes)
Type of Service (ToS) / DSCP	8 bits	Defines priority/QoS handling: minimize delay, maximize throughput, maximize reliability, minimize cost
Total Length	16 bits	Total length of the datagram (header + data) in bytes. Max = 65,535 bytes
Identification	16 bits	Uniquely identifies a datagram; all fragments of the same datagram share this value
Flags	3 bits	Bit 0: reserved. Bit 1: DF (Don't Fragment) . Bit 2: MF (More Fragments) — 1 means more fragments follow, 0 means this is the last
Fragment Offset	13 bits	Position of this fragment's data in the original datagram, measured in units of 8 bytes
Time To Live (TTL)	8 bits	Maximum number of hops. Decrement by each router; when it reaches 0 the packet is discarded and an ICMP message is sent. Prevents infinite loops
Protocol	8 bits	Identifies the upper-layer protocol: 1 = ICMP, 2 = IGMP, 6 = TCP, 17 = UDP, 89 = OSPF
Header Checksum	16 bits	Error check on the header only (not data). Recomputed at each router because TTL changes
Source Address	32 bits	IP address of the sender
Destination Address	32 bits	IP address of the receiver
Options	0–40 bytes	Optional: Record Route, Strict/Loose Source Route, Timestamp, No Operation, End of Option

4.3 Fragmentation

Different networks have different **MTU (Maximum Transmission Unit)** values — the largest frame size the data link layer can carry.

Network	MTU
Ethernet	1500 bytes
PPP	296 bytes
FDDI	4352 bytes
Token Ring (16 Mbps)	17,914 bytes

When a datagram is larger than the outgoing link's MTU, the router **fragments** it.

Rules: - Fields that **change** during fragmentation: Total Length, Flags (MF), Fragment Offset, Header Checksum. - Fields **copied** into every fragment: Identification, Source, Destination. - Fragment offset is expressed in units of **8 bytes**, so every fragment's data (except the last) must be a multiple of 8. - **Reassembly is done only at the final destination**, not at intermediate routers.

Worked Fragmentation Example

A datagram of **4000 bytes** (20-byte header + 3980 bytes of data) must cross a link with **MTU = 1500**.

Each fragment can hold at most $1500 - 20 = 1480$ bytes of data (a multiple of 8 ✓).

Fragment	Data bytes	Offset (bytes)	Offset field (+8)	MF flag	Total Length
1	1480	0	0	1	1500
2	1480	1480	185	1	1500
3	1020	2960	370	0	1040

4.4 Supporting Protocols at the Network Layer

ARP — Address Resolution Protocol

Maps a **known IP address** to an **unknown MAC address**.

Process: 1. The host broadcasts an **ARP Request** frame: "Who has IP 192.168.1.5? Tell 192.168.1.2." 2. All hosts on the LAN receive it, but only the one owning that IP replies. 3. That host sends a **unicast ARP Reply** containing its MAC address. 4. The result is stored in the **ARP cache** for a short time to avoid repeat queries.

ARP variations: - **Proxy ARP** — a router answers ARP requests on behalf of hosts on another network. - **Gratuitous ARP** — a host announces its own IP-MAC mapping (used to detect duplicate IPs).

RARP — Reverse ARP

Maps a **known MAC address** to an **unknown IP address**. Used by diskless workstations at boot. Now obsolete — replaced by **BOOTP** and **DHCP**.

ICMP — Internet Control Message Protocol

IP provides no error reporting; ICMP fills that gap. ICMP messages are **encapsulated inside IP datagrams** (Protocol = 1).

Error-reporting messages: | Type | Message | When sent | |---|---|---| | 3 | Destination Unreachable | Router/host cannot deliver the datagram | | 4 | Source Quench | (Deprecated) Congestion control — slow down | | 11 | Time Exceeded | TTL reached 0, or reassembly timer expired | | 12 | Parameter Problem | Ambiguous or missing header field value | | 5 | Redirection | Informs a host of a better route |

Query messages: | Type | Message | Use | |---|---|---| | 8 / 0 | Echo Request / Echo Reply | Used by **ping** to check reachability | | 13 / 14 | Timestamp Request / Reply | Round-trip time measurement |

Tools using ICMP: - **ping** — checks if a host is alive and measures RTT using Echo Request/Reply. - **traceroute / tracert** — discovers the route by sending packets with increasing TTL values (1, 2, 3...) and collecting the Time Exceeded messages from each router.

IGMP — Internet Group Management Protocol

Manages **multicast group membership**. Hosts use it to tell their local router they want to join or leave a multicast group. Message types: Query, Membership Report, Leave Report.

4.5 Routing — Fundamentals

Routing is the process of selecting a path for traffic across one or more networks. A **router** forwards packets based on its **routing table**.

A Routing Table Contains

- **Destination network / mask**
- **Next-hop address** (or "directly connected")
- **Outgoing interface**
- **Metric / cost**
- **Flags** (U = up, G = gateway, H = host-specific)

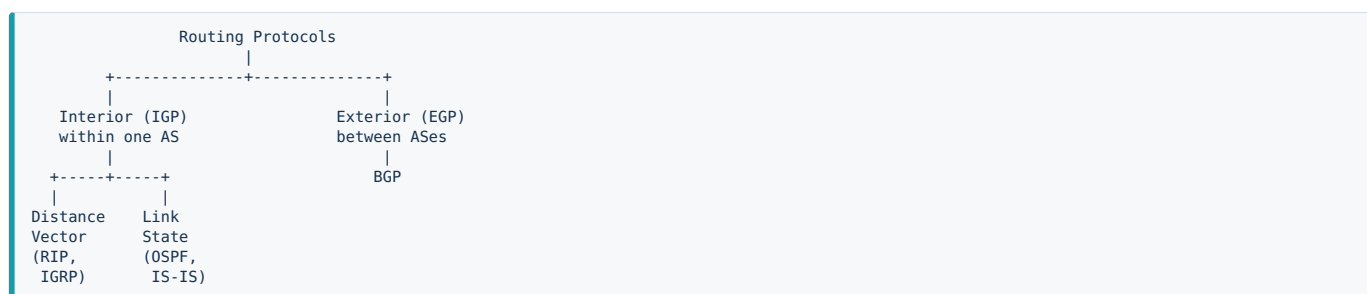
Types of Routing

Type	Description
Static Routing	Routes configured manually by an administrator. Simple, secure, no overhead — but doesn't adapt to failures. Good for small/stub networks
Dynamic Routing	Routers exchange information and update routes automatically using routing protocols. Adapts to topology changes; more overhead
Default Routing	A single route (<code>0.0.0.0/0</code>) used when no specific match exists — the "gateway of last resort"

Forwarding Techniques

- **Next-hop method** — the table only holds the address of the next hop, not the full route.
- **Network-specific method** — one entry for an entire network rather than one per host.
- **Host-specific method** — an entry for a particular destination host (used for special cases).
- **Default method** — one entry covering everything else.
- **Longest Prefix Match** — when multiple entries match, the router chooses the one with the **longest (most specific) prefix**.

Routing Protocol Classification



An **Autonomous System (AS)** is a group of networks and routers under a single administrative authority.

4.6 Distance Vector Routing

Each router maintains a **vector (table)** of the least cost to every destination and shares it with its **direct neighbors** periodically.

Principle — Bellman-Ford Algorithm

$D(x, y) = \min \{ c(x, v) + D(v, y) \}$ for each neighbor v

Each router says: "To reach Y , my best cost is my cost to neighbor V plus V 's cost to Y — pick the minimum."

Characteristics

- Routers know only about their **neighbors**, not the full topology ("routing by rumor").
- Updates are sent **periodically** (e.g., RIP every 30 seconds) and when changes occur.
- Simple to implement; low memory and CPU needs.

Problems

Count-to-Infinity: When a link fails, bad information can circulate, with routers incrementing each other's costs slowly toward infinity.

Solutions: 1. **Defining Infinity** — treat 16 as infinity (RIP), so the loop ends quickly. 2. **Split Horizon** — never advertise a route back out the interface it was learned from. 3. **Split Horizon with Poison Reverse** — advertise it back, but with a metric of infinity. 4. **Hold-down timers** — ignore updates about a route for a period after it goes down. 5. **Triggered updates** — send an update immediately on a topology change instead of waiting for the timer.

4.7 RIP — Routing Information Protocol

An interior, distance-vector protocol.

Feature	Detail
Metric	Hop count (each network = 1 hop)
Maximum hop count	15 (16 = infinity/unreachable)
Update interval	Every 30 seconds
Transport	UDP port 520
Algorithm	Bellman-Ford
Versions	RIPv1 (classful, broadcast), RIPv2 (classless/VLSM support, multicast to 224.0.0.9, authentication), RIPng (for IPv6)

Timers: Periodic (30s), Expiration (180s), Garbage Collection (120s).

Limitations: Small maximum network size, slow convergence, metric ignores bandwidth and delay.

4.8 Link State Routing

Every router builds a **complete map of the entire network topology**, then independently computes the shortest path.

Process

1. Each router **discovers its neighbors** and the cost to each (via Hello packets).
2. It builds an **LSP (Link State Packet)** containing node identity, list of neighbors, costs, sequence number, and age.
3. LSPs are **flooded** to every other router in the area.
4. Each router assembles a **Link State Database (LSDB)** — identical on all routers.
5. Each router runs **Dijkstra's Shortest Path First algorithm** on the LSDB to build a shortest-path tree rooted at itself, then derives its routing table.

Dijkstra's Algorithm — Steps

1. Set the root node's cost to 0 and all others to infinity; mark all as tentative.
2. Move the tentative node with the lowest cost into the permanent set.
3. Update the costs of that node's neighbors if a cheaper path is found through it.
4. Repeat until all nodes are permanent.

Distance Vector vs Link State

Aspect	Distance Vector	Link State
Knowledge	Only about neighbors	Full network topology
Shares with	Only direct neighbors	All routers (flooding)
Shares what	Its entire routing table	Only link-state info about its own links
Update frequency	Periodic	Event-triggered
Algorithm	Bellman-Ford	Dijkstra (SPF)
Convergence	Slow	Fast
Loop risk	High (count-to-infinity)	Low
CPU / memory	Low	High
Examples	RIP, IGRP	OSPF, IS-IS

4.9 OSPF — Open Shortest Path First

An interior, link-state protocol; the most common IGP in large enterprise networks.

Feature	Detail
Algorithm	Dijkstra's SPF
Metric	Cost , typically based on bandwidth (Cost = 10^8 / bandwidth in bps)
Transport	Directly over IP, Protocol number 89
Multicast addresses	224.0.0.5 (all OSPF routers), 224.0.0.6 (all DRs)
Classless	Yes — supports VLSM and CIDR
Hierarchy	Divides an AS into areas ; Area 0 is the backbone

Router types: Internal Router, Area Border Router (ABR), Backbone Router, AS Boundary Router (ASBR).

Packet types: Hello, Database Description (DD), Link State Request, Link State Update, Link State Acknowledgement.

Link types in OSPF: Point-to-point, Transient (multi-access, e.g. Ethernet), Stub, Virtual.

Why areas? They limit the scope of LSA flooding and reduce LSDB size and SPF computation load.

4.10 BGP — Border Gateway Protocol

The **exterior gateway protocol** that runs the Internet — it routes between Autonomous Systems.

Feature	Detail
Type	Path vector protocol
Transport	TCP port 179 (reliable)
Current version	BGP-4
Metric	Not a simple metric — uses path attributes and policy

Path Vector: Instead of just a cost, BGP advertises the complete **AS_PATH** (the list of ASes a route passes through). This: - Prevents loops (a router rejects a path already containing its own AS number). - Enables **policy-based routing** (political, security, and economic decisions, not just shortest path).

Types: - **eBGP (External BGP)** — between routers in different ASes. - **iBGP (Internal BGP)** — between BGP routers within the same AS.

Message types: Open, Update, Keepalive, Notification.

Key path attributes: AS_PATH, NEXT_HOP, LOCAL_PREF, MED (Multi-Exit Discriminator), ORIGIN, WEIGHT (Cisco-specific).

4.11 Multicast Routing — Brief

- **Unicast** — one source to one destination.
- **Multicast** — one source to a group of destinations.
- **Broadcast** — one source to all hosts on a network.

Multicast routing protocols: DVMRP (distance-vector based), MOSPF (extension of OSPF), PIM-DM (Dense Mode — flood and prune), PIM-SM (Sparse Mode — uses a Rendezvous Point), CBT (Core-Based Trees).

Trees used: Source-Based Tree (shortest path tree per source) and Group-Shared Tree (a single tree per group, rooted at a core/RP).